TMA3084 SOFTWARE ENGINEERING LAB

Annabell, 102439

PROPOSAL

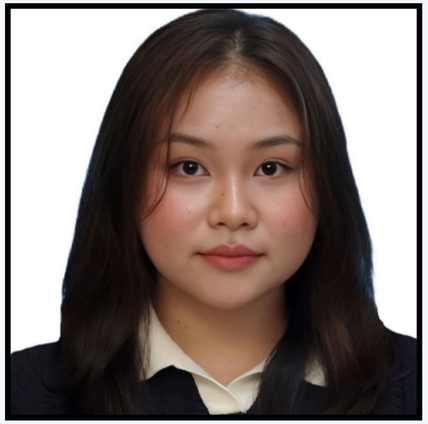
PRESENTATION

nägel : Nail Business Management System

Group: NailMuse, G02_T29

OUR TEAM

01 AIDA NAZURA BINTI MOHAMED AMIN
103132



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01

INTRODUCTION

PROJECT & CLIENT

- nägel app is a Flutter-based mobile application designed for a home-based boutique nail salon.

OPERATIONAL MOTIVATION

- Transitions salon management from manual, chat-based channels (WhatsApp/Instagram) to a centralized mobile booking platform.

CORE VALUE PROPOSITION

- Appointment scheduling
- Deposit tracking
- Inventory management
- OpenAI-powered virtual assistant for automated FAQs and personalized design recommendations.

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02 CLIENT CONTEXT

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THE BUSINESS

CLIENT	• Ms. Michelle Anak Ini • nägel, home-based boutique nail salon • Batu Kawa, Kuching
BUSINESS MODEL	• Appointment-based salon offering manicures, custom gel extensions, and nail care.
CURRENT OPERATING ENVIRONMENT	• Managing all bookings, deposit verifications, and reminders manually over WhatsApp and Instagram.

TARGET USER PERSONAS

01

New Customer:
Needs: View service catalogue, transparent prices, and available booking slots without waiting for long chat replies.

02

Regular Customer:
Needs: Book far in advance, save design reference photos/preferences, and receive automated appointment notifications.

03

Salon Owner (Ms. Michelle):
Needs: Single dashboard to set calendar availability, check deposit receipts, review design preferences beforehand, and track material stock.

TEAM SETUP & ROLES

Annabell (Software Project Manager)	**Victoria** (Lead Systems Analyst & Requirements Engineer)	**Aida** (Lead Mobile Architect (Flutter/Dart))	**Yolanda** (Backend & AI Integration Specialist)	**Bruce** (Quality Assurance & CI/CD Engineer)
• Team Leader • Customer Interface • FR-01, FR-06, FR-11	• Planning Manager • FR-02, FR-03, FR-12	• Design & Implementation Manager • FR-07, FR-09, FR-14	• Process & Support Manager • FR-08, FR-10, FR-15, FR-16	• Quality & Test Manager • FR-04, FR-05, FR-13

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PROBLEM STATEMENT

CHALLENGES FACED BY nägel

P1

Booking & Cancellations

Bookings mainly through WhatsApp; last minute cancellations disrupt scheduling.

P2

Changing Nail Designs

Design changes during appointments make appointment difficult to estimate.

P3

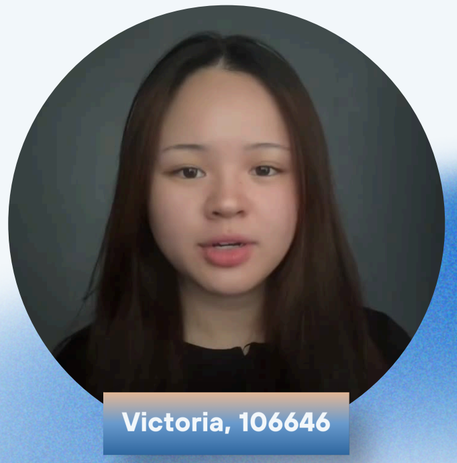
Manual Reminder

The owner sometimes forgets to remind customers before appointments.

P4

Stock & Expenses

No systematic tracking method; spending can exceed the budget when purchasing supplies for cutsomer requests.



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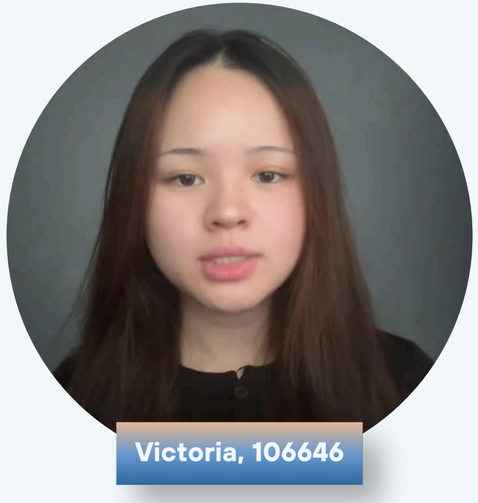
PROPOSED SOLUTION

NAIL BUSINESS MANAGEMENT SYSTEM (NBMS)



Booking & Scheduling
Manage appointments and view available time slots.

Customer Reminders
Support appointment reminders and notifications.

Stock & Expenses
Record supplies and monitor business spending.

AI-powered Assistance
Recommend nail designs, estimate prices and answer FAQs.

Expected benefits

- More organised appointment management
- Better nail design preparation before appointment
- More convenient customer access to service information
- Improved visibility of stock and expenses

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TARGET USER FLOW

CUSTOMER: FROM INSTAGRAM TO A BOOKED SLOT

1. Open link in Instagram bio
2. Choose service and see price
3. Pick an open date and time
4. Details and photos
5. Pay deposit
6. Booking confirmed

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TARGET USER FLOW

06

OWNER: FROM BOOKING TO APPOINTMENT

A Set open days and times

B See bookings and photos

C Check the deposit receipt

OWNER HOURS: MON–FRI FROM 7:00 PM · SAT–SUN 10:00 AM – 7:30 PM

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MOBILE UX

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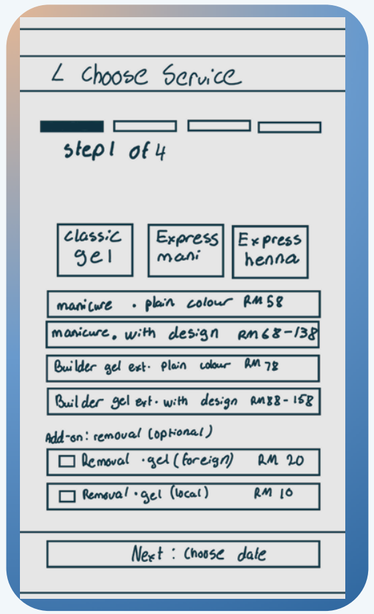
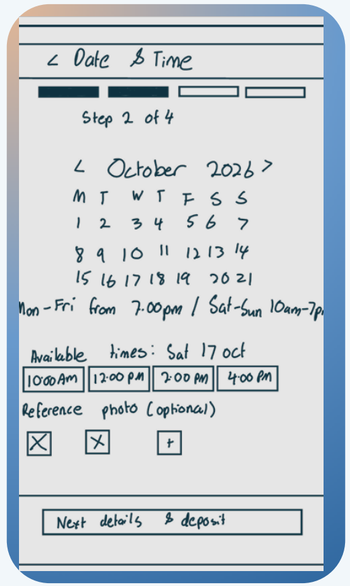
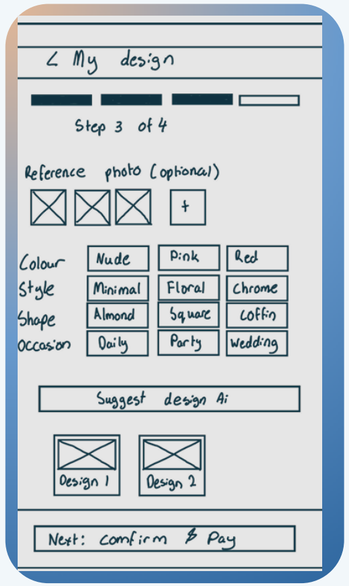
Screen 1 – Choose Service



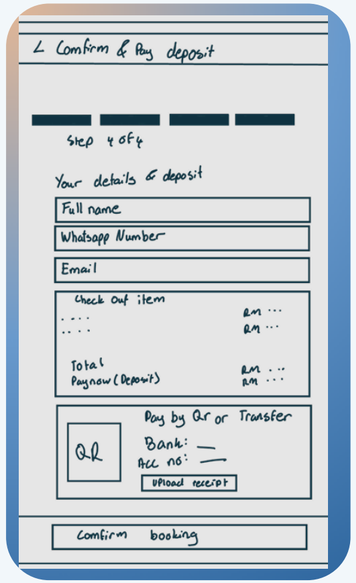
Screen 2 – Date, & Time



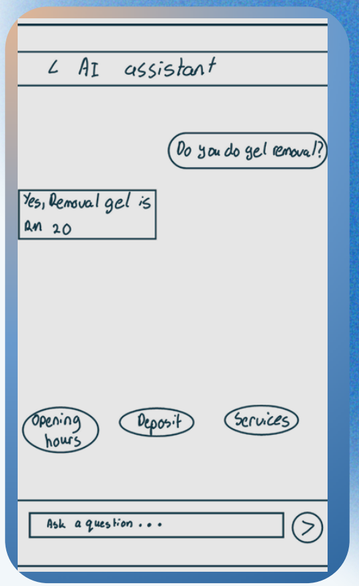
Screen 3 – MY Design (AI)



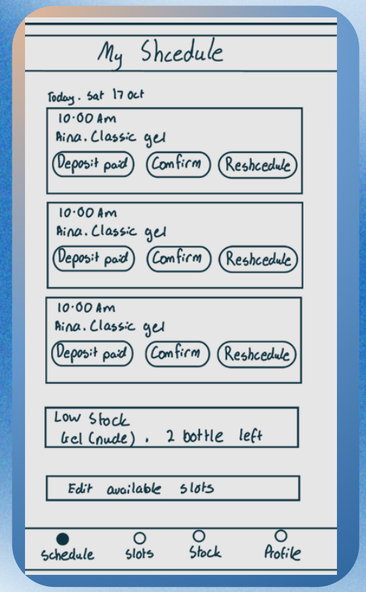
Screen 4 – Confirm and Deposits



Screen 5– AI assistant



Screen 6 – Owner: My schedule



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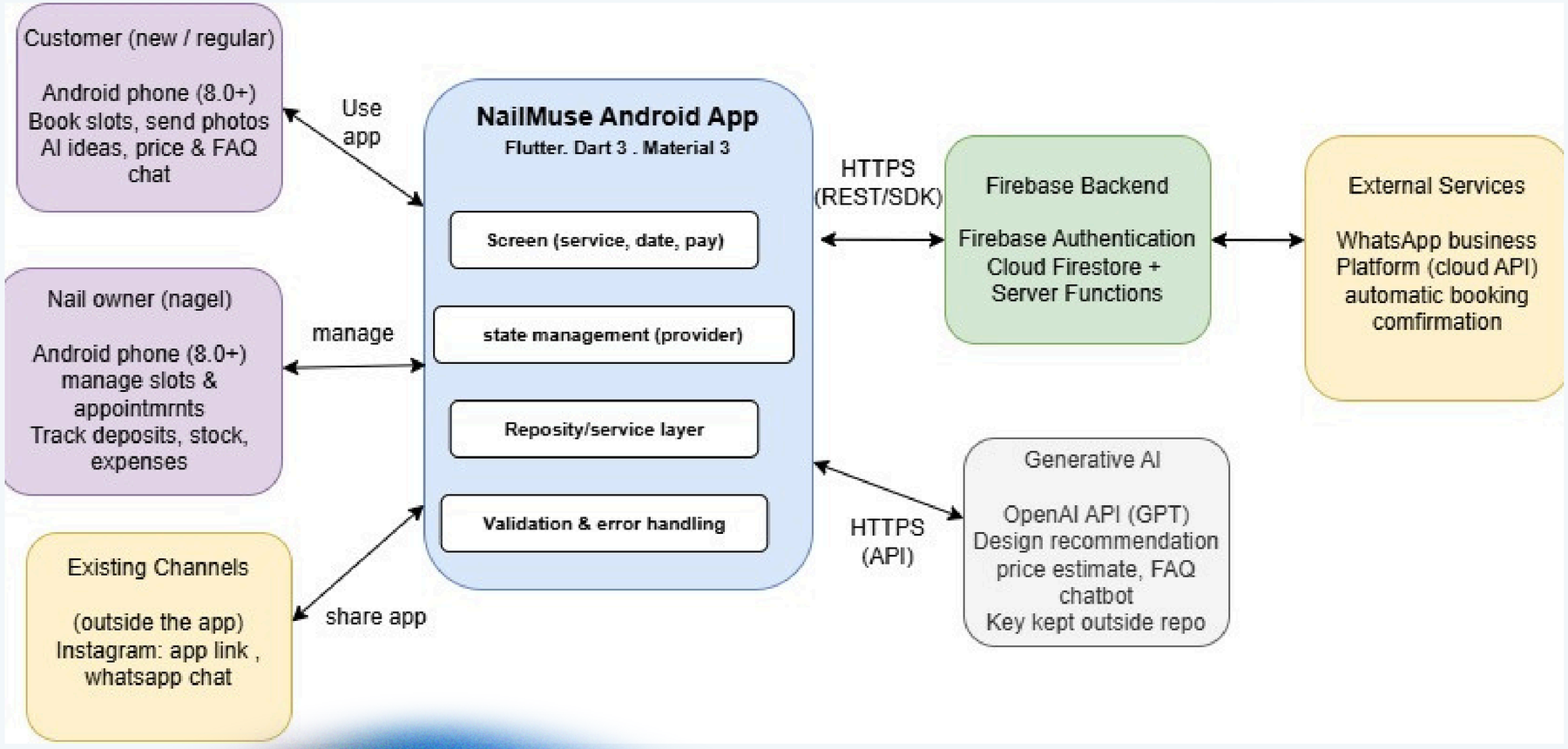
SYSTEM ARCHITECTURE



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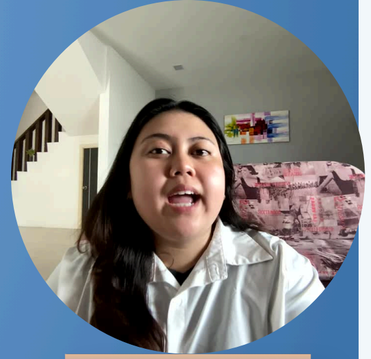

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🤖 OPENAI GENERATIVE AI ENGINE

AI Nail Design Recommendation

- Customers enter preferences such as:
  - Nail style
  - Colour preferences
  - Occasion (wedding, graduation, festive, etc.)
  - Budget
  - Reference images
- AI recommends suitable nail designs based on the inputs.

AI Virtual Assistant (Chatbot)

- Answers frequently asked questions (FAQs)
- Provides information about:
  - Services offered
  - Pricing
  - Appointment booking
  - Nail care & aftercare
- Available 24/7 for customer enquiries.

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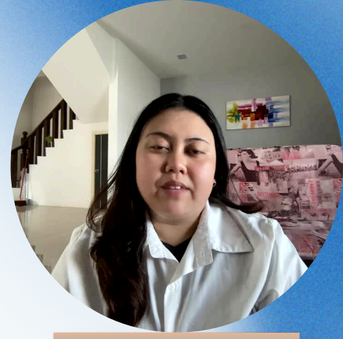
NON-FUNCTIONAL REQUIREMENTS



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💡 KEY NON-FUNCTIONAL REQUIREMENTS (NFRS)

NFR-ID	Category	Target
NFR-01	Performance	App screens load within 3 seconds
NFR-02	Security	No API keys exposed in repository
NFR-03	Availability	99% system availability
NFR-04	Usability	Booking process completed within 5 minutes
NFR-05	Compatibility	Android 8.0 and above
NFR-06	AI Response Time	AI responses delivered within 5 seconds

AI SECURITY CONSTRAINTS

API KEY PROTECTION

- OpenAI API keys are stored securely and never harcoded into the source code.
- Keys are excluded from public GitHub repositories.

DATA PRIVACY

- Only necessary customer information is sent to the AI service.
- Sensitive business and customer data remain stored in Firebase.

INTERNET DEPENDENCY

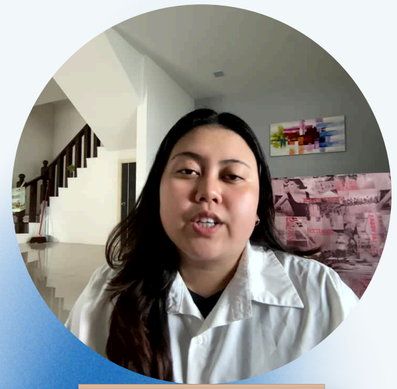
- AI features require an internet connection to function.
- Core features remain available even if AI services are unavailable.

COST MANAGEMENT

- OpenAI API usage generates costs based on request volume.
- Usage must be monitored to prevent exceeding the project budget.

RESPONSE RELIABILITY

- AI recommendations may vary and should be treated as suggestions rather than final professional advice.



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PROJECT ROADMAP

M1
Scope & Requirements
Weeks 4-5

- GitHub repository
- Issue board & CI
- Proposal & SRS

M2
Design & Prototype
Week 9

- SDS draft
- 4–5 screens
- Green CI & v0.2.0

M3
Product & Verification
Week 13

- At least 50 tests
- At least 60% coverage
- Working AI feature

M4
Final Release & Acceptance
Week 15

- Release APK v1.0.0
- Client sign-off
- e-ArchivE

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RISK MANAGEMENT

RISK 1
Client Availability

Late client feedback may delay development and acceptance testing.

Mitigation

Schedule check-ins early and record client feedback and decisions.

RISK 2
AI API Limitations

API usage limits, costs or downtime may affect AI recommendations.

Mitigation

Monitor usage, protect API credentials and keep the service catalogue accessible if AI is unavailable.

RISK 3
Technical Skill Gaps

Limited experience with Flutter, backend integration or testing may cause delays.

Mitigation

Share knowledge, consult documentation and test features incrementally.

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CONCLUSION

SUMMARY OF CORE VISION

The nägel mobile app transforms manual, chat-based salon operations into a modern, automated mobile solution.

KEY BUSINESS & USER IMPACT

- **For Customers:** Seamless booking, clear pricing, saved preferences, and 24/7 AI-powered assistance.
- **For Salon Owner (Ms. Michelle):** Reduced inquiry workload, organized deposit tracking, automated reminders, and supply expense visibility.

COMMITMENT TO QUALITY

Built on a robust TSPi framework, clean Flutter/Firebase architecture, and automated CI test coverage (≥ 50% test cases, ≥ 60\% code coverage).

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THANK YOU

INSPIRED BY YOU,
DESIGNED FOR YOU.

NailMuse.